COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 8 : SLO 2 — Restoring vs Non-Restoring Division
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

---

Section C: Simulate the Algorithm (Apply): 0100 (4) ÷ 0010 (2) — non-restoring

• Init: A = 0000, Q = 0100, M = 0010
• Step 1: shift → A = 0000, Q = 1000; A = A - M = 1110 (< 0), Q0 = 0 → A = 1110, Q = 1000
• Step 2: shift → A = 1101, Q = 0000; A < 0 → A = A + M = 1111 (< 0), Q0 = 0 → A = 1111, Q = 0000
• Step 3: shift → A = 1110, Q = 0000; A < 0 → A = A + M = 0000 (≥ 0), Q0 = 1 → A = 0000, Q = 0001
• Step 4: shift → A = 0000, Q = 0010; A ≥ 0 → A = A - M = 1110 (< 0), Q0 = 0 → A = 1110, Q = 0010
• Final correction: A = A + M = 0000. Quotient = 0010 (2), Remainder = 0000 (0) ✓

Section D: Debrief — comparing logic flow and efficiency

• Efficiency: non-restoring removes the restore step, so every cycle costs exactly one addition/subtraction plus a shift — fewer operations and simpler control than restoring division.
• Correction at the end: a negative final A is not a valid remainder, so the divisor is added back once to fix it.
• Quotient-bit assignment: restoring sets the bit only after a possible restore; non-restoring sets it straight from the sign of A (1 if A ≥ 0, 0 if A < 0).
• Logic flow: restoring = shift → subtract → if negative restore (add back) → set bit; non-restoring = shift → add or subtract depending on the sign of A → set bit → single correction at the end.